

ROI Study - Slope Revelation - Confidence in False Statements (#246653)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has one author.

Pre-registered on:

2025/09/11 13:05 (PT)

1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

In a cue-learning task in which none of the cues has an influence on the outcome, people who learned in the presence of a positive (vs. negative) exogenous trend will be more likely to believe that the (non-influential) cues had an impact on the outcome.

3) Describe the key dependent variable(s) specifying how they will be measured.

DV1: Subjective understanding. Participants will report their subjective understanding of the relationship between cues and outcomes on three 5-point bipolar scale items. These will be averaged into a composite measure of subjective understanding.

DV2: Confidence in correctness of false statements. Participants will see three statements, each claiming that a non-diagnostic cue had an impact on the outcome. They will be asked to report their belief that these false statements are correct on 5-point scale items anchored at Not at all confident (1) and Extremely confident (5). These three items will be averaged into a composite measure of confidence.

4) How many and which conditions will participants be assigned to?

Participants will be randomly assigned to one of two cells of a 2 (Slope: Increasing vs. Flat) design.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

We will analyze each of the two dependent variables using two sample t-tests, comparing the score on the DV between the "Increasing" and "Flat" condition.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We will measure the amount of time participants spend on the cue-learning task, and exclude all participants who completed the task in less than 90 seconds.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

Following our Pre-Registered Interim Analysis Design (PRIAD; André and Reinholtz 2024), we will collect up to 800 observations and perform 1 interim analysis.

We will test for significance at $N = [400, 800]$.

Following the Pocock (1977) correction, we will reject the null and stop data collection if p is respectively smaller than $[0.02939, 0.02939]$ at each of these stages. If, at the interim analysis, only one of our two DVs is significant, we will proceed to collect the second batch of data.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We will measure, at the end of the experiment, whether participants believed that the outcomes increased (vs. stayed flat) over time. We will report this variable as a descriptive outcome.